

Asante Twi Text-to-Speech: Fine-tuning VITS on Open Data

[Your name / student ID] · MPhil Project · Variety: Asante Twi (ISO 639-3: aka) · Code: [repo link] · Samples & checkpoint: [folder link]

1. Data and Preprocessing

****Variety.**** Asante Twi, chosen because it has the largest amount of openly available speech and because its orthography is nearly phonemic, which suits a grapheme-based front end.

****Dataset.**** [Name, URL, licence (e.g. CC-BY 4.0), speaker count, recording conditions]. A single speaker ([speaker id, gender]) was selected to avoid speaker-identity variance in a small-data setting. Total raw audio: [x.x h]; after filtering: [x.x h, N utterances, train/val split]. Justification: [why this corpus is suitable: studio/clean audio, sentence-level transcripts, licence permits derivative models].

****Preprocessing**** (scripts/prepare_data.py): resampling to 22.05 kHz mono; silence trimming (top_db=35) with 100 ms padding; peak normalisation to -1 dBFS; duration filter 1-15 s; a characters-per-second sanity filter (4-25) to remove misaligned pairs; random validation split. Utterances dropped: [n dur / n text / n rate].

****Text normalisation**** (scripts/normalize.py): Unicode NFC; mapping of look-alike characters (Greek ε/Ε, Ε, Ϛ) to ε and Ϛ; lowercasing; removal of combining tone marks; expansion of digits 0-99 into Twi number words (larger numbers read digit by digit); restriction to a 28-letter + punctuation inventory. ****No G2P**** was used: Twi spelling maps closely to sounds, and ε/Ϛ are separate model symbols, so the model learns the open-vowel contrast directly. Final symbol set: [paste charset from stats.json].

2. Modelling Methodology

****Architecture.**** VITS (Kim et al., 2021): a conditional VAE with normalising-flow prior, stochastic duration predictor and a HiFi-GAN-style decoder trained adversarially. Because VITS is end-to-end, the acoustic model and vocoder are trained jointly and no separate vocoder is needed. Chosen over two-stage pipelines (FastPitch + HiFi-GAN) because it needs less tuning, has fewer failure points under a short deadline, and works well from small datasets when fine-tuned.

****Fine-tuning.**** Initialised from the pretrained English LJSpeech VITS checkpoint (Coqui model zoo). Only the text-embedding layer (new Akan symbol inventory) is re-initialised; all other weights are transferred, since low-level acoustic and decoder knowledge is language-independent. Key hyperparameters: lr (gen/disc) 1e-4, AdamW, batch size [16], mixed precision [on/off], epochs/steps [300 / N steps], 22.05 kHz, 80 mel bins, hop 256. Best checkpoint chosen by [validation loss / listening]. Environment: [Colab T4 16 GB], training time [x h].

Component	Source
Reused	coqui-tts (VITS implementation, Trainer), pretrained LJSpeech VITS weights, MMS-1b-all ASR (evaluation only), librosa, jiwer
Own work	Twi text normaliser and number expansion; data cleaning/filtering pipeline; training configuration and character set; inference wrapper; ASR-based and listener evaluation tooling; test set

3. Evaluation and Results

****Test set.**** Ten sentences not in the training data (data/test_sentences.txt), covering ε/Ϛ minimal contexts, numbers, questions, and long sentences. Native-speaker checked: [yes/no].

****Objective.**** ASR round trip: synthesised audio is transcribed with MMS-1b-all (adapter aka) and scored by WER/CER against the input text. Limitation: the score includes ASR errors, so the same ASR was also run on real held-out recordings as a reference ceiling; it measures intelligibility only, not tone or naturalness, and MMS accuracy on Twi is itself limited.

Condition	WER	CER	n
Synthesised (test set)	[[]]	[[]]	10
Real held-out speech (ceiling)	[[]]	[[]]	[[]]

****Listener evaluation.**** [N] Twi-speaking listeners rated each sample for naturalness and intelligibility (1-5) and flagged mispronounced words, ε/Ϛ quality and tone/prosody problems (eval/mos_sheet.csv). Results: [MOS naturalness x.x ± CI, intelligibility x.x ± CI; inter-rater notes]. With so few raters the figures are indicative, not statistically robust.

4. Discussion and Limitations

****Orthography and ε/Ϛ.**** Because ε and Ϛ are separate graphemes, the model can in principle learn their open-vowel quality; failure modes to check are collapse towards e/o (influence of the English initialisation) and inconsistent rendering in unstressed syllables. Findings: [observed behaviour]. Datasets often contain inconsistent spellings (e.g. e/ε, o/Ϛ typed with ASCII), which directly corrupts supervision, hence the normalisation step.

****Vowel harmony.**** Akan has ATR vowel harmony (+ATR i e u o a vs -ATR ɪ ɛ ʊ ɔ ɑ); the orthography writes ɪ and ʊ as i and u, and does not mark the harmonic class of a. A grapheme model must infer the ATR value from context, so errors are likely on affixes (e.g. pronoun and tense prefixes) that change quality with the root vowel. A phoneme-level front end with explicit ATR marking is a clear improvement. Observed: [\[examples\]](#).

****Tone.**** Akan is a two-level (H/L) tone language with downstep; tone is lexically and grammatically contrastive but unmarked in standard writing, so the model sees only ambiguous graphemes and learns an average prosody. Expected errors: flattened or English-like intonation, wrong tone on homographs and on tense/aspect distinctions. Observed: [\[examples\]](#).

****Other limitations.**** Small single-speaker data limits robustness to rare words, names, loanwords and numbers above 99; possible artefacts: [\[buzz, repeated or skipped syllables, breaths, long-sentence drift\]](#); the number words and test items were [\[verified by a native speaker\]](#).

****Improvements.**** (1) Add tone- and ATR-annotated lexicon or a rule-based G2P; (2) pretrain on multilingual or Niger-Congo data (e.g. MMS-TTS, other Akan varieties) before fine-tuning; (3) more speakers with careful single-speaker adaptation; (4) a proper MOS study with more raters and tone-pair minimal-pair tests; (5) data augmentation by text-selection for tonal minimal pairs.

References

Kim, J., Kong, J., Son, J. (2021). Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech. ICML.
Kong, J. et al. (2020). HiFi-GAN. NeurIPS. · Pratap, V. et al. (2023). Scaling Speech Technology to 1,000+ Languages (MMS). · Coqui TTS, github.com/idiap/coqui-ai-TTS. · [\[Dataset citation\]](#)